# EARLY WARNING INDICATORS OF FINANCIAL CRISIS

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# Literature on Banking Crises



**Banking crises**

| No. | Source | Coverage and definition |
|---|---|---|
| 1. | Caprio and Klingebiel (2003) | The annual dataset (1970–2002) includes information on 117 episodes of systemic banking crises in 93 countries and on 51 episodes of borderline and non-systemic banking crises in 45 countries.<br><br>A systemic crisis is defined as 'much or all of bank capital was exhausted.' Expert judgment was also employed 'for countries lacking data on the size of the capital losses, but also for countries where official estimates understate the problem.' |
| 2. | Kaminsky and Reinhart (1999) | The monthly dataset (1970–1995) includes 26 episodes of banking crisis in 20 countries.<br><br>Banking crises are defined by two types of events: '(1) bank runs that lead to the closure, merging, or takeover by the public sector of one or more financial institutions; and (2) if there are no runs, the closure, merging, takeover, or large-scale government assistance of an important financial institution (or group of institutions) that marks the start of a string of similar outcomes for other financial institutions.'<br><br>The dataset of banking crises was compiled using existing studies of banking crises and the financial press. |
| 3. | Laeven and Valencia (2008, 2010, 2012) | The annual dataset (1970–2011) covers systemically important banking crises (147 episodes) in over 100 countries all over the world and provides information on crisis management strategies.<br><br>A banking crisis is considered to be systemic if the following two conditions are met: '(1) Significant signs of financial distress in the banking system (as indicated by significant bank runs, losses in the banking system, and/or bank liquidations); and (2) Significant banking policy intervention measures in response to significant losses in the banking system.' The first year that both criteria are met is considered to be the starting year of the banking crisis, and policy interventions in the banking sector are considered significant if at least three out of the following six measures were used: '(1) extensive liquidity support; (2) bank restructuring costs; (3) significant bank nationalizations; (4) significant guarantees put in place; (5) significant asset purchases; and (6) deposit freezes and bank holidays.'<br><br>The dataset is compiled using the authors' calculations combined with some elements of judgment for borderline cases. |
| 4. | Reinhart and Rogoff (2008, 2011) | The annual dataset (1800–2010, from the year of independence) covers banking crises in 70 countries.<br><br>The definition of banking crisis is the same as in Kaminsky and Reinhart (1999) (see above).<br><br>The dataset of banking crises was compiled using existing studies of banking crises and the financial press. |

# Literature on Sovereign Debt Crises



**Debt crises**

| No. | Source | Definition and coverage |
|---|---|---|
| 1. | Detragiache and Spilimbergo (2001) | The annual dataset (1971–1998) includes 54 episodes of debt crisis in 69 countries.<br><br>A debt crisis is defined as a situation when 'either or both of following conditions occur: (1) there are arrears of principal or interest on external obligations towards commercial creditors (banks or bondholders) of more than 5 percent of total commercial debt outstanding; (2) there is a rescheduling or debt restructuring agreement with commercial creditors as listed in Global Development Finance (World Bank). The 5 percent minimum threshold is to rule out cases in which the share of debt in default is negligible, while the second criterion is to include countries that are not technically in arrears because they reschedule or restructure their debt obligations before defaulting.' |
| 2. | Laeven and Valencia (2008, 2010, 2012) | The annual dataset (1970–2011) includes 66 episodes of debt crisis in over 100 countries all over the world.<br><br>Sovereign debt default and restructuring episodes are dated on the basis of various studies, including reports from the IMF, the World Bank and rating agencies. |
| 3. | Levy-Yeyati and Panizza (2011) | The annual dataset (1970–2005) includes 63 episodes of debt crisis in 39 countries.<br><br>The dataset is compiled by the authors using Standard & Poor's, the World Bank's Global Development Finance database (analysis and statistical appendix), and press reports. |
| 4. | Reinhard and Rogoff (2011) | The annual dataset (1800–2010, from the year of independence) tracks episodes of both external and domestic debt crises in 70 countries.<br><br>An 'external debt crisis involves outright *default* on payment of debt obligation incurred under foreign legal jurisdiction, including nonpayment, *repudiation*, or the *restructuring* of debt into terms less favorable to the lender than in the original contract.' A domestic debt crisis incorporates the definition of external debt crisis and, in addition, the freezing of bank deposits and/or forcible conversion of foreign currency deposits into local currency. |

# Literature on Currency Crises



Currency (balance of payment) crises

| No. | Source | Definition and coverage |
|---|---|---|
| 1. | Kaminsky and Reinhart (1999) | The monthly dataset (1970–1995) includes 76 episodes of currency crisis in 20 countries.<br><br>A currency crisis is defined excessive exchange rate volatility ('turbulence'), that is, when the index representing a weighted average of changes in the exchange rate and reserves exceeds a certain threshold. 'Crisis episodes' are then defined as 'the month of the crisis plus the 24 months preceding the crisis.' For a robustness check, two alternative windows are used, starting at 12 and 18 months prior to the crisis.<br><br>The dataset is compiled using the authors' calculations. |
| 2. | Kaminsky (2006) | The monthly dataset (1970–2002) includes 96 episodes of currency crisis in 20 industrial and developing countries.<br><br>The definition of currency crises and 'crisis episodes' is as in Kaminsky and Reinhart (1999).<br><br>The dataset is compiled using the authors' calculations. |
| 3. | Laeven and Valencia (2008, 2010, 2012) | The annual dataset (1970–2011) includes 218 currency crises identified in over 100 countries all over the world.<br><br>A currency crisis is defined as 'a nominal depreciation of the currency vis-à-vis the U.S. dollar of at least 30 percent that is also at least 10 percentage points higher than the rate of depreciation in the year before… For countries that meet the criteria for several continuous years, we use the first year of each 5-year window to identify the crisis.' It should be noted that this list also includes large devaluations by countries that adopt fixed exchange rate regimes. |
| 4. | Reinhart and Rogoff (2011) | The annual dataset (1800–2010, from the year of independence) tracks currency crises (also called 'crashes') in 70 countries.<br><br>A currency crisis is defined as an excessive exchange rate depreciation, that is, when the annual depreciation vis-à-vis USD or the relevant anchoring currency (GBP, FRF, DM, EUR) exceeds the threshold value of 15%.<br><br>The dataset is compiled using the authors' calculations. |

# Some Literature on Financial Crises



| Study | Sample and Frequency | Country Coverage | Indicators | Comments |
|---|---|---|---|---|
| **Kaminsky and Leiderman (1995)** | 1985–1987, monthly | Argentina, Israel, and Mexico | (1) monetary shocks<br>(2) fiscal shocks<br>(3) past inflation | Discuss the probability of crisis in exchange-rate-based stabilization programs. |
| **Kaminsky and Reinhart (1996)** | 1970–1995, monthly | 20 countries; 5 industrial and 15 developing; 76 currency crises and 26 banking crises | (1) export growth<br>(2) import growth<br>(3) bilateral real exchange rate—deviation from trend<br>(4) terms of trade changes<br>(5) changes in reserves<br>(6) money demand/supply gap<br>(7) changes in bank deposits<br>(8) real interest rates<br>(9) lending-deposit spread<br>(10) domestic-foreign real interest rate differential<br>(11) M2 money multiplier<br>(12) M2/international reserves<br>(13) growth in domestic credit/GDP<br>(14) changes in stock prices<br>(15) output growth<br>(16) financial liberalization<br>(17) banking crises | The behavior of (1)-(15) is examined 18 months before and after the crises and compared to the behavior of these indicators during "tranquil" periods. (16-17) are used in predicting the **probability** of crises. The usefulness of all the indicators is assessed by: (a) determining whether they gave a **signal on a** crisis by crisis basis; (b) tabulating the probability of crisis conditioned on a signal from the individual indicator; and (c) tabulating the probability of false signals. |
| **Klein and Marion (1994)** | 1957–1991, monthly | 87 peg episodes, as in Flood and Marion (1995). | (1) bilateral real exchange rates<br>(2) real exchange rate squared<br>(3) net foreign assets of the monetary sector/M1<br>(4) net foreign assets of the monetary sector/M1 squared<br>(5) openness<br>(6) trade concentration<br>(7) regular executive transfers<br>(8) irregular executive transfers<br>(9) months spent in the peg | Using pooled data (1)-(8) are used to estimate the probability of devaluation at t +1; the sample is disaggregated into pre- and post-Bretton Woods period and distinctions are made between pegs that are followed by either a float or a crawling peg from devaluations followed by a new peg. |
| **Krugman (1996)** | 1988–1995, annual, quarterly, some daily | France, Italy, Spain, Sweden, and the United Kingdom during the 1992–93 ERM crises. | (1) unemployment rate<br>(2) output gap<br>(3) inflation<br>(4) public debt/GDP | While the bulk of the paper is theoretical, evidence on the trends of (1)-(4) is used to support the argument that the ERM episode does not provide evidence of self-fulfilling crises. |

Note: Additional details on the individual countries included in the larger cross-country studies are available in the original studies.

# Some Literature on Financial Crises



| Study | Sample and Frequency | Country Coverage | Indicators | Comments |
|---|---|---|---|---|
| **Milesi-Ferretti and Razin (1995)** | 1970– 94, annual | Chile and Mexico have the 4 crises cases; Ireland, Israel, and South Korea are no crises cases due to policy reversal; and Australia no crisis case with no policy change. | (1) debt service/GDP adjusted for GDP growth and changes in the real exchange rate<br>(2) exports/GDP<br>(3) real exchange rate versus historical norm<br>(4) saving/GDP<br>(5) fiscal stance<br>(6) fragility of the banking sector<br>(7) political instability<br>(8) composition of capital flows | The emphasis is on developing a notion of current account sustainability and the factors it depends on. While there is no formal test, (1)-(8) are used to compare the crises and no crises episodes. |
| **Moreno (1995)** | 1980–94, monthly and quarterly | Indonesia, Japan, Malaysia, Philippines, Singapore, Korea, and Thailand. 126 episodes of speculative pressures; 72 in the direction of depreciation and 54 in the direction of appreciation. | (1) change in bilateral exchange rate<br>(2) changes in net foreign assets (central bank)<br>(3) domestic-foreign interest rate differential<br>(4) exports/imports<br>(5) output gap<br>All the following are relative to the United States:<br>(6) growth of domestic credit/reserve money<br>(7) growth in M1<br>(8) growth in broad money<br>(9) fiscal deficit/government spending<br>(10) inflation | The emphasis is on testing whether the behavior of macroeconomic variables (4)-(10) differs between "tranquil" and "speculative" periods. (1)-(3) are used to define such periods. |
| **Otker and Pazarbasioglu (1994)** | 1979–93, monthly | Denmark, Ireland, Norway, Spain, and Sweden. The sample covers 15 devaluations and 10 realignments of all central rats. | (1) domestic credit<br>(2) real effective exchange rate<br>(3) trade balance<br>(4) unemployment rate<br>(5) German price level<br>(6) output<br>(7) reserves<br>(8) central parity<br>(9) foreign-domestic interest rate differential<br>(10) position within band | The aim is to use (1)-(10) to estimate the probability of abandoning the peg by either devaluing or floating. (1)-(8) are associated for macroeconomic "fundamentals" while (7), (9), and (10) are proxies for "speculative factors." |

Note: Additional details on the individual countries included in the larger cross-country studies are available in the original studies.

# Some Literature on Financial Crises



| Study | Sample and Frequency | Country Coverage | Indicators | Comments |
|---|---|---|---|---|
| **Otker and Pazarbasioglu (1995)** | 1982–1994, monthly | Mexico. During the sample there are 4 devaluations; 3 increases in the rate of crawl, and 2 reductions; and 2 shifts to a more flexible exchange system. | (1) real exchange rate<br>(2) international reserves<br>(3) inflation differential with the United States<br>(4) output growth<br>(5) U.S. interest rates<br>(6) central bank credit to the banking system<br>(7) financial sector reform dummy<br>(8) share of short-term foreign currency debt<br>(9) fiscal deficit<br>(10) current account balance | (1)-(10) are used to estimate the one-step ahead probability of a regime change. The relative importance of the indicators is assessed for pre- and post-November 1991, when the dual exchange rate system was abandoned. |
| **Sachs, Tornell, and Velasco (1995)** | 1985–1995, monthly and annual | 20 emerging market countries | (1) the real exchange rate<br>(2) credit to the private sector/GDP<br>(3) M2/international reserves<br>(4) saving/GDP<br>(5) investment/GDP<br>(6) capital inflows/GDP<br>(7) short-term capital inflows/GDP<br>(8) government consumption/GDP<br>(9) current account/GDP | The emphasis is on explaining why some countries were more affected by the Mexican crisis than others. |

Note: Additional details on the individual countries included in the larger cross-country studies are available in the original studies.

# Plausible Leading Indicators of Banking Crises… (1)



| | | | | |
|---|---|---|---|---|
| 1 | baaspread | BAA corporate bond spread | none | Reuters |
| 2 | capform | Gross total fixed capital formation (constant prices) | % qoq | Statistical offices, OECD |
| 3 | comprice | Commodity prices | % qoq | Commodity Research Bureau |
| 4 | curaccount | Current account (%GDP) | none | OECD, WDI |
| 5 | domprivcredit | Domestic credit to private sector (%GDP) | none | WDI |
| 6 | fdiinflow | FDI net inflows (%GDP) | none | WDI |
| 7 | govtcons | Government consumption (constant prices) | % qoq | OECD, statistical offices |
| 8 | govtdebt | Government debt (%GDP) | none | WDI, ECB |
| 9 | hhcons | Private final consumption expenditure (constant prices) | % qoq | Statistical offices |
| 10 | hhdebt | Gross liabilities of personal sector | % qoq | National central banks, Oxford Economics |
| 11 | houseprices | House price index | % qoq | BIS, Eurostat, Global Property Guide |
| 12 | indprodch | Industrial production index | % qoq | Statistical offices |
| 13 | indshare | Industry share (%GDP) | none | WDI, EIU |
| 14 | inflation | Consumer price index | % qoq | Statistical offices, national central banks |
| 15 | m1 | M1 | % qoq | National central banks |
| 16 | m3 | M3 | % qoq | National central banks |

- Several variables may be related to financial crisis but for finding which ones might have a leading indicators property for giving an early warning on financial crisis is a statistical issue that needs to settled through empirical research
- Various macro-financial variables/ indicators that may be considered may be divided into: (i) leading indicators, (ii) coincident indicators, (iii) lagging indicators and (iv) not related to crisis
- Ex-post episodes of stress can be divided into binary scores of 1 for crisis (banking/debt/ currency) and 0 for otherwise
- In practice useful to track the following indicators:
  - private credit boom: Domestic private credit growing far above its long-term trend (often 2% or more above baseline).
  - Loan-to-deposit ratio: Rising fast, showing banks are lending more money than they safely take in from deposits.
  - Short-term debt reliance: Heavy use of risky, short-term wholesale funding to pay for long-term loans.

# Plausible Leading Indicators of Banking Crises… (2)



| | | | | |
|---|---|---|---|---|
| 17 | mmrate | Money market interest rate | none | IFS |
| 18 | neer | Nominal effective exchange rate | % qoq | IFS |
| 19 | netsavings | Net national savings (%GNI) | none | WDI |
| 20 | shareprice | Stock market index | % qoq | Reuters, stock exchanges |
| 21 | taxburden | Total tax burden (%GDP) | none | OECD, statistical offices |
| 22 | termsoftrade | Terms of trade | none | Statistical offices |
| 23 | trade | Trade (%GDP) | none | WDI |
| 24 | trbalance | Trade balance | 1st dif | Statistical offices, national central banks |
| 25 | wcreditpriv | Global domestic credit to private sector (%GDP) | none | WDI |
| 26 | wfdiinflow | Global FDI inflow (%GDP) | none | WDI |
| 27 | winf | Global inflation | none | IFS |
| 28 | wrgdp | Global GDP | % qoq | IFS |
| 29 | wtrade | Global trade (constant prices) | % qoq | IFS |
| 30 | yieldcurve | Long term bond yield – money market interest rate | none | National central banks |

| | | |
|---|---|---|
| (i) | Banking | Banking crises (1 if a crisis was reported, 0 otherwise) |
| (ii) | Debt | Debt crises (1 if a crisis was reported, 0 otherwise) |
| (iii) | Currency | Currency crises (1 if a crisis was reported, 0 otherwise) |

- BIS monitors credit-to-GDP gap as an important early warning indicator for potential banking crises or severe distress. It is a notion of "excessive credit" calculated in a simple way, captures credit as total borrowing from all domestic and foreign sources. The credit-to-GDP gap is defined as the difference between the credit-to-GDP ratio and its long-run trend. The trend is derived using a one-sided (ie backward-looking) Hodrick-Prescott filter.
- HP filter finds a trend component that minimizes the variance of the cyclical component while simultaneously penalizing sudden changes in the trend's growth rate:

The formula for the minimization problem is:

$$\min \sum_{t=1}^{T} (y_t - \tau_t)^2 + \lambda \sum_{t=1}^{T} [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2$$

Where:

- $y_t$ is the original time series data.
- $\tau_t$ is the smoothed trend component.
- $y_t - \tau_t$ represents the cyclical component (the gap).
- λ is the smoothing parameter that dictates the sensitivity of the trend.

# Leading Indicators of Banking Crises



| | (LPM, FE) banking_onset4q | (LOGIT, FE) banking_onset4q | (RELOGIT) banking_onset4q |
|---|---|---|---|
| main | | | |
| domprivcredit | 0.00113*** | 0.0153*** | 0.0108*** |
| | (9.04) | (5.59) | (6.64) |
| fdiinflow | 0.00415*** | 0.0416*** | 0.00571* |
| | (5.04) | (2.75) | (1.69) |
| winf | 0.00614*** | 0.101*** | 0.0746*** |
| | (9.17) | (7.82) | (7.52) |
| mmrate | 0.00159*** | 0.0196*** | 0.0206*** |
| | (4.82) | (3.45) | (4.81) |
| wrgdp | 0.00597** | 0.172*** | 0.228*** |
| | (2.57) | (2.79) | (4.57) |
| trbalance | -5.03e-08*** | -0.00000143*** | -0.00000107*** |
| | (-3.00) | (-2.86) | (-3.72) |
| trade | 0.00105*** | 0.0216** | 0.0113*** |
| | (2.75) | (2.26) | (4.69) |
| reer | 0.383*** | 9.847*** | 8.927*** |
| | (3.60) | (3.98) | (4.06) |
| govtbalance | 0.00510*** | 0.121*** | 0.0717*** |
| | (4.53) | (3.99) | (3.99) |
| _cons | -0.239*** | | -6.691*** |
| | (-7.55) | | (-16.00) |
| *N* | 3377 | 3047 | 3377 |

Note: 1. LPM, FE – linear probability model (panel fixed effects estimator), 2. LOGIT, FE – limited dependent variable model (panel logit fixed effects estimator), and 3. RELOGIT – limited dependent variable model for rare events (pooled logit), t statistics in parentheses, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In ECB's research work, the following variables are found to have strong leading indicator properties:

- Domestic private credit
- FDI inflow
- World inflation
- Money market interest rates
- World GDP
- Trade balance
- Trade as a % of GDP
- Real Effective Exchange Rate
- General government balance

***BUT to my Mind the best Leading Indicator of a banking crises would be a stress in money market with some entities undertaking trades at off-market rates***

# Plausible Indicators of Currency Crises



- **Capital account:** international reserves, capital flows, short-term capital flows, foreign direct investment, and the differential between domestic and foreign interest rates.
- **Debt profile:** public foreign debt, total foreign debt, short-term debt, share of debt classified by type of creditor and by interest structure, debt service, and foreign aid.
- **Current account:** the real exchange rate, the current account balance, the trade balance, exports, imports, the terms of trade, the price of exports, savings and investment.
- **International variables**: foreign real GDP growth, interest rates, and price level.
- **Financial liberalization:** credit growth, the change in the money multiplier, real interest rates, and the spread between bank lending and deposit interest rates.
- **Other financial variables:** central bank credit to the banking system, the gap between money demand and supply, money growth, bond yields, domestic inflation, the "shadow" exchange rate, the parallel market exchange rate premium, the central exchange rate parity, the position of the exchange rate within the official band, and M2/international reserves.
- **Real sector:** real GDP growth, the output gap, employment/unemployment, wages, and changes in stock prices.
- **Fiscal variables:** the fiscal deficit, government consumption, and credit to the public sector.
- **Institutional/structural factors:** openness, trade concentration, and dummies for multiple exchange rates, exchange controls, duration of the fixed exchange rate periods, financial liberalization, banking crises, past foreign exchange market crises, and past foreign exchange market events.[16]
- **Political variables:** dummies for elections, incumbent electoral victory or loss, change of government, legal executive transfer, illegal executive transfer, left-wing government, and new finance minister; also, degree of political instability (qualitative variable based on judgement).

# Leading Indicators of Currency Crises



| Sector | Variables | Number of Studies Considered | Statistically Significant Results |
|---|---|---|---|
| Capital account | international reserves | 11 | 10 |
| | short-term capital flows | 2 | 1 |
| | foreign direct investment | 1 | 1 |
| | capital account balance | 1 | -- |
| | domestic-foreign interest differential | 2 | 1 |
| Debt profile | foreign aid | 1 | -- |
| | external debt | 1 | -- |
| | public debt | 1 | 1 |
| | share of commercial bank loans | 1 | -- |
| | share of concessional loans | 1 | 1 |
| | share of variable-rate debt | 1 | -- |
| | share of short-term debt | 2 | -- |
| | share of multilateral development bank debt | 1 | -- |
| Current account | real exchange rate | 12 | 10 |
| | current account balance | 6 | 2 |
| | trade balance | 3 | 2 |
| | exports | 3 | 2 |
| | imports 1/ | 2 | 1 |
| | terms of trade | 2 | 1 |
| | export prices | 1 | -- |
| | savings | 1 | -- |
| | investment | 1 | -- |
| International | foreign real GDP growth | 1 | -- |
| | foreign interest rates | 3 | 1 |
| | foreign price level | 2 | 1 |
| Financial liberalization | real interest rates | 1 | 1 |
| | credit growth | 7 | 5 |
| | lending-deficit interest spread | 1 | -- |
| | money multiplier | 1 | 1 |
| Other financial | parallel market premium | 1 | 1 |
| | central parity | 1 | 1 |
| | position within the band | 1 | 1 |
| | money demand-supply gap | 1 | 1 |
| | change in bank deposits | 1 | -- |
| | central bank credit to banks | 1 | 1 |
| | money | 3 | 2 |
| | M2/international reserves | 2 | 2 |
| Real sector | inflation 2/ | 5 | 5 |
| | real GDP growth or level | 8 | 5 |
| | output gap | 1 | 1 |
| | employment/unemployment 3/ | 3 | 2 |
| | change in stock prices | 1 | 1 |

In Kaminsky, Lizondo and Reinhart's research work, the following variables are found to have strong leading indicator properties:

- International Reserves
- Real exchange rate
- Credit growth
- Inflation
- GDP growth or level or output gap
- Money supply
- M2 / international reserves
- Short-term debt
- FDI
- Domestic-foreign interest rate differential
- Public debt
- Employment/ unemployment
- Change in stock prices